ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

NATURAL SCIENCES

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 7 Natural Sciences paper 1 · practice d.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 C ✓ 1.3 A ✓ 1.4 D ✓ 1.5 A ✓ 1.6 C ✓ 1.7 B ✓ 1.8 D ✓ 1.9 A ✓ 1.10 C ✓ (10)

QUESTION 2

2.1 True ✓✓ (2)

2.2 False ✓ – six ✓ (2)

2.3 True ✓✓ (2)

2.4 False ✓ – good ✓ (2)

2.5 True ✓✓ (2)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3

3.1 Lithosphere – rocks, soil, mountains ✓; hydrosphere – oceans, rivers, ice ✓; atmosphere – air, oxygen, clouds ✓; biosphere – all living things, plants and animals ✓ (4)

3.2 Any TWO ✓✓: vertebrates have a backbone (spine), invertebrates do not; vertebrates have an internal skeleton, many invertebrates have an exoskeleton or a soft body; most vertebrates are larger (2)

3.3 (a) Birds ✓ (b) Reptiles ✓ (c) Mammals ✓ (3)

3.4 So that the young plants do not compete with the parent plant (and each other) ✓ for light, water, space and nutrients, and so the plant can spread to new areas ✓. (2)

3.5 (a) Wind ✓ – it is light and the parachute lets it float ✓ (b) Animals ✓ – animals eat the fruit and drop the seeds elsewhere in their droppings ✓ (4)

3.6.1 The differences ✓ between individuals of the same species ✓ (2)

3.6.2 Any ONE ✓: eye colour; hair type; skin colour; shoe size; mass; tongue rolling; blood group (1)

3.6.3 When conditions change (e.g. a new disease or climate change), some individuals have features that help them survive ✓, so the species does not die out ✓. (2)

[20]

QUESTION 4

4.1.1 P: liquid ✓ Q: solid ✓ R: gas ✓ (3)

4.1.2 Q ✓ – it has a very high melting point, so it will not melt on a stove ✓ (2)

4.2 Evaporation: heat the salty water (or leave it in the sun) ✓ until all the water evaporates ✓; the water turns into a gas and leaves, but the salt does not evaporate and stays behind as crystals ✓. (3)

4.3.1 Pink ✓ (1)

4.3.2 Green ✓ (1)

4.3.3 So that the solutions do not mix and give a wrong result (fair test) ✓ (1)

4.4 (a) When an acid and a base react ✓ and cancel each other out to form a neutral solution (salt and water) ✓ (b) Heartburn is caused by too much stomach acid ✓; the antacid is a base that neutralises the acid ✓ (4)

4.5 Any THREE ✓✓✓: good conductor of electricity; ductile (can be pulled into thin wires); flexible / malleable; does not rust easily; strong (3)

4.6 On the right-hand side ✓ (1)

4.7 Any ONE ✓: oxygen for breathing / hospitals; carbon as graphite in pencils; chlorine to clean swimming-pool water; nitrogen in fertilisers; helium in balloons (1)

[20]

QUESTION 5

5.1 (a) Conduction ✓ (b) Convection ✓ (c) Radiation ✓ (3)

5.2 (a) Heat cannot be conducted or convected through a vacuum ✓ (b) The shiny walls reflect heat radiation back into the flask ✓ (c) It stops hot air escaping by convection (and is made of an insulator) ✓ (3)

5.3 Heat (thermal) energy ✓ ... electrical energy ✓ (2)

5.4 Advantages, any TWO ✓✓: South Africa has a lot of coal; reliable power day and night; creates jobs; cheaper to produce in large amounts. Disadvantages, any TWO ✓✓: non-renewable; air pollution and acid rain; carbon dioxide adds to global warming; mining damages the land; uses a lot of water. (4)

5.5 Sunlight will not run out ✓; it is replaced naturally every day ✓. (2)

5.6 Any FOUR ✓✓✓✓: switch off lights and appliances when not in use; use LED bulbs; boil only the water you need; use a solar geyser or geyser blanket; switch the geyser off when not needed; unplug chargers; use natural light; dry clothes outside (4)

5.7 Any TWO ✓✓: people can be electrocuted and die; fires can start; the grid can be overloaded, causing power cuts; children can touch exposed live wires (2)

[20]

QUESTION 6

6.1 Any TWO ✓✓: it is a star; it is a ball of hot gases (hydrogen and helium); it is the centre of the solar system; it gives the Earth light and heat; it is about 150 million km from Earth; it is much bigger than the Earth (2)

6.2 (a) About 24 hours (one day) ✓ (b) About 365¼ days (one year) ✓ (2)

6.3 Sun, Earth and Moon in a straight line ✓ with the Earth in the middle ✓; the Moon in the Earth's shadow, all labelled ✓ (3)

6.4 The Sun's light is so strong that it can permanently damage your eyes ✓ and cause blindness ✓. (2)

6.5 Geocentric: the Earth is at the centre ✓ and the Sun, Moon and planets move around it ✓. Heliocentric: the Sun is at the centre ✓ and the Earth and planets move around the Sun ✓. (4)

6.6 Nicolaus Copernicus ✓ (accept Aristarchus) (1)

6.7 Any TWO ✓✓: they saw moons around Jupiter; craters and mountains on the Moon; more stars than can be seen with the naked eye; phases of Venus, which supported the heliocentric model; new planets were discovered (2)

6.8 The Moon does not make its own light ✓; it reflects light from the Sun ✓. (2)

6.9 Any correct example with a short explanation ✓✓, e.g. the Pleiades (isiLimela) appearing at dawn in winter told farmers that it was time to prepare to plough and plant; the Milky Way seen as a path across the sky in San stories; the Khoikhoi linked the Pleiades to the start of the rainy season (2)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE